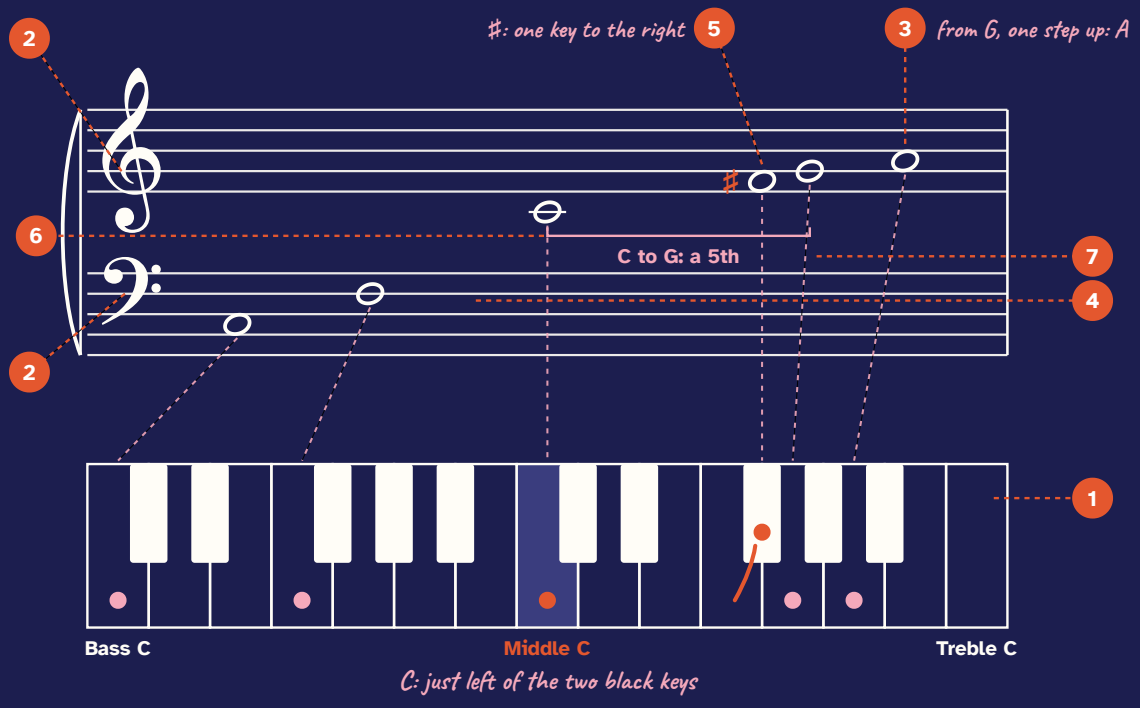
PART I · WHICH NOTE

Pitch

High or low? That is the first question every note answers. Start with the keys under your fingers, find where they live on the staff, see how sharps and flats move them, then measure how far apart two notes are. Everything in this part is in this one picture.



1 **The keyboard** p. 22
two and three black keys; C is left of the two

2 **The staff and the clefs** p. 26
the treble clef names G, the bass clef F

3 **Reading the notes** p. 29
start from a clef note, then step or skip

4 **The staff and the keyboard** p. 32
every line and space is one white key

5 **Sharps, flats and naturals** p. 35
a sharp moves one key right, a flat one key left

6 **Counting intervals** p. 38
count the notes, both ends: C to G is a 5th

7 **Major, minor and perfect** p. 41
count the keys between: 6 make a perfect 5th

★ **Tips: when a note reads wrong** p. 44

PART I · PITCH · 1

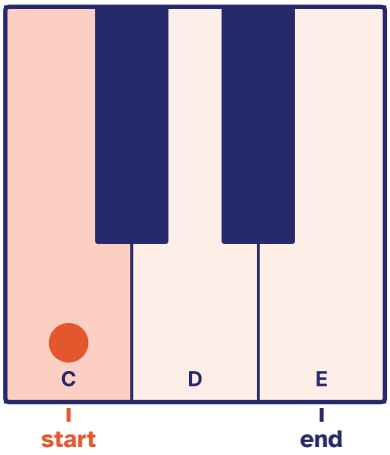
The keyboard

Left is **low**, right is **high**. The black keys come in groups of **two** and **three**. Each group has a first white key, just to its left: **C** for the two, **F** for the three. Find the group, find its first key, then walk on.

THE SECRET

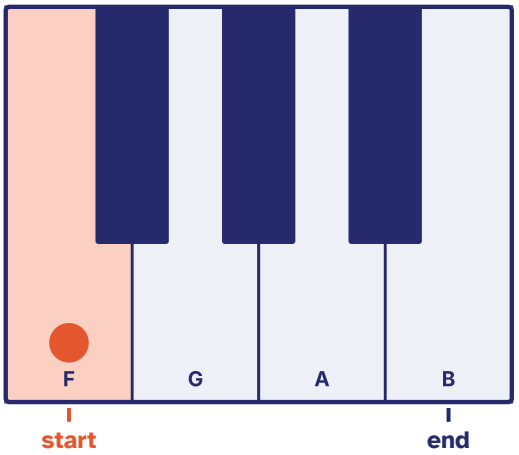
Two black keys: start on C. Three black keys: start on F. Then walk on.

2 black keys: C D E



on the piano E and F touch

3 black keys: F G A B



Pulled apart: C starts the two, F starts the three · E and B end them

THE RULE

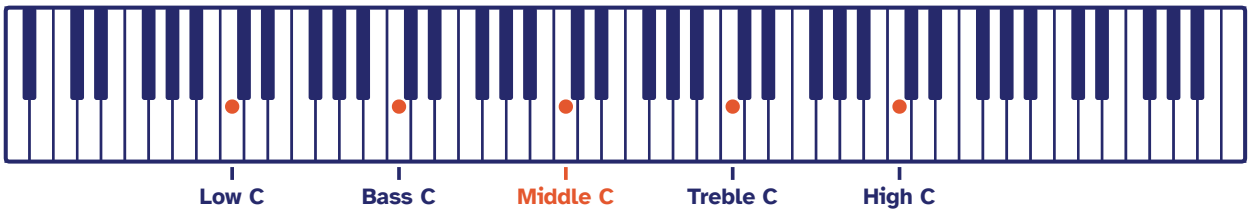
Find the group, then its first key. Looking for D or E? Find two black keys and start on C. Looking for G, A or B? Find three black keys and start on F. You never have to count from far away.

Seven letters, two families

The white keys use seven letters, **A B C D E F G**, then start again, like the days of the week. They live in two families: **C D E** round the two black keys, **F G A B** round the three.

The five Cs have names

The C nearest the middle is **Middle C**. The Cs either side are **Bass C** and **Treble C**; further out, **Low C** and **High C**. From one C to the next is an **octave**: C D E F G A B C, eight notes, both Cs included.

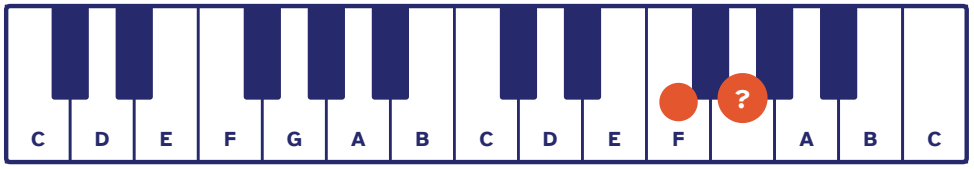


The whole piano, all 88 keys: the five named Cs sit in the middle

PART I · PITCH · 1 · CONTINUED

EXAMPLE 1

Find G

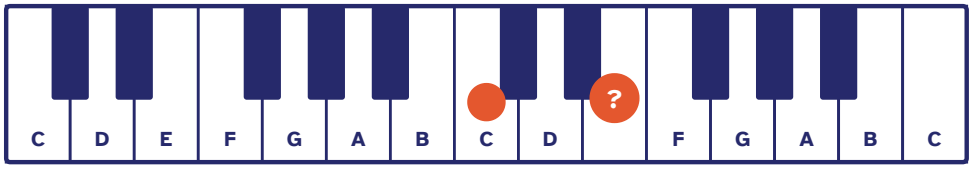


1. **Which family?** G is in F G A B: look for **three** black keys.
2. **Find its first key:** F, just left of the three (the red dot).
3. **Walk on one key.** That is the **?**.

The **?** is **G**: the key after F. No need to start from C and count five.

EXAMPLE 2

Find E



1. **Which family?** E is in C D E: look for **two** black keys.
2. **Find its first key:** C, just left of the two (the red dot).
3. **Walk on:** C, D … the third key is the **?**.

The **?** is **E**: the last of the C D E family, just right of the two black keys.

EXAMPLE 3

Find B



1. **Which family?** B is in F G A B: three black keys.
2. **Start on F** (the red dot).
3. **Walk on:** F, G, A … the last key of the family is the **?**.

The **?** is **B**: the last of F G A B, just right of the three black keys.

GAME

Which family?

1	2	3	4	5	6
D	A	F	E	B	G

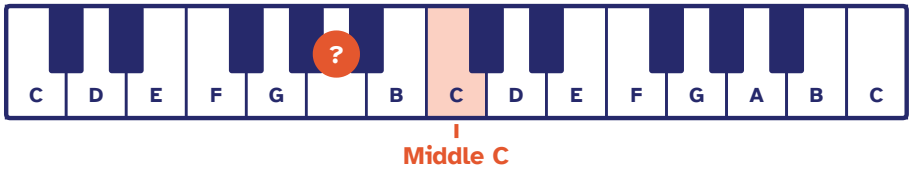
Does each white key live with the **two** black keys (start on C) or the **three** (start on F)?

Answer: 1 two · 2 three · 3 three (the first) · 4 two (the last) · 5 three (the last) · 6 three.

PART I · PITCH · 1 · CONTINUED

EXAMPLE 4

Find the A below Middle C

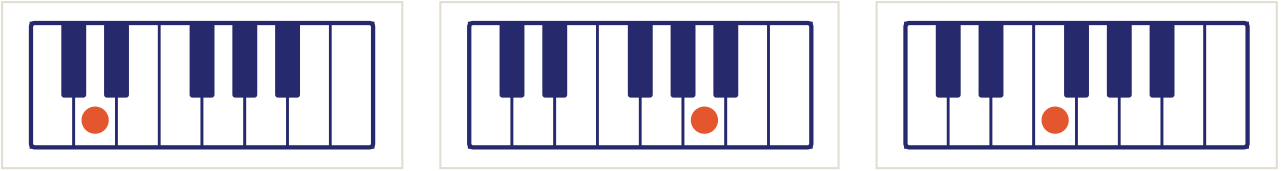


1. **Which family?** A is in F G A B.
2. **From Middle C, look left** for the nearest three black keys, and start on their F.
3. **Walk on:** F, G … the third key is the **?**.

The **?** is **A**: F, G, A. Find the family first, and every key is at most three steps away.

GAME

Name the key



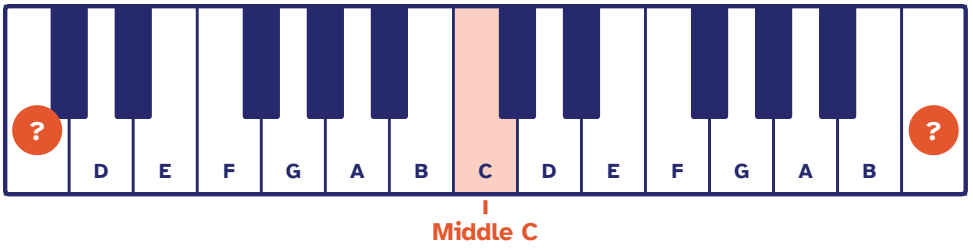
1 2 3

Find the group, then its first key, then walk on. Name each red key.

Answer: 1 D (C, then D) · 2 A (F, G, A) · 3 F (the first of the three).

EXAMPLE 5

Find Bass C and Treble C



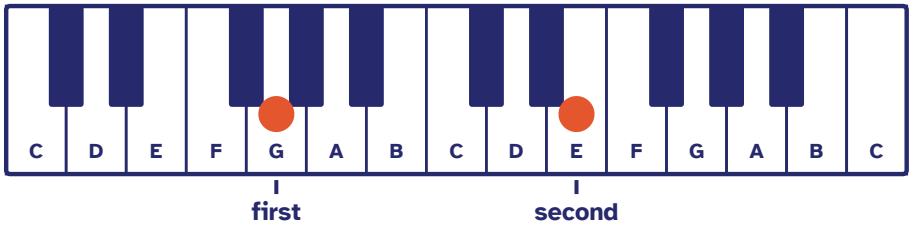
1. **Start on Middle C.**
2. **Walk left** to the next pair of black keys. The C just left of it is one **?**.
3. **Walk right** the same way, to the next pair: the other **?**.

Left **?**: **Bass C**. Right **?**: **Treble C**. Each is one octave, eight notes, from Middle C.

PART I · PITCH · 1 · CONTINUED

EXAMPLE 6

Higher or lower?



1. **Find the two red keys.**
2. **Which one is further right?**
3. **Right means higher:** play both and listen.

The second note is **higher**: it is to the right. Left is always lower, right always higher.

TRY IT · 10 SECONDS

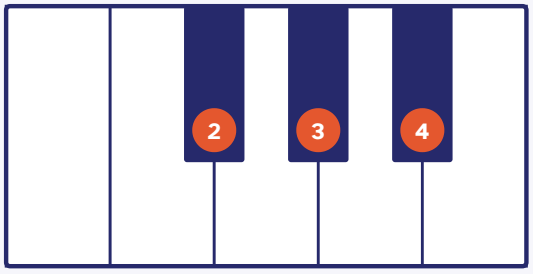


Play every C, then every F.

Bottom to top, saying the names: Low C, Bass C, Middle C, Treble C, High C. Then every F: left of the three.

YOU KNOW THIS

Hot Cross Buns, on three black keys



Right hand, fingers **2 3 4** on the three black keys. Play from the top: **4 3 2**, 4 3 2, then 2 2 2 2, 3 3 3 3, 4 3 2. A whole tune, and not one white key.

Try it. Find another group of three, higher up, and play it again there.

THINK OF IT The black keys are bumps you can feel in the dark. Close your eyes and slide your hand along: two bumps, and the key to their left is C; three bumps, and the key to their left is F.

TRAP

- × Counting every key from C: find the nearest group and start from its C or F.
- × Mixing up the families: C is left of the **two** black keys, F left of the **three**.

SEE ALSO

The staff and the clefs p. 26 · The staff and the keyboard p. 32

PRACTISE IT

Workbook Pitch 1 · Online: Library › Keyboard

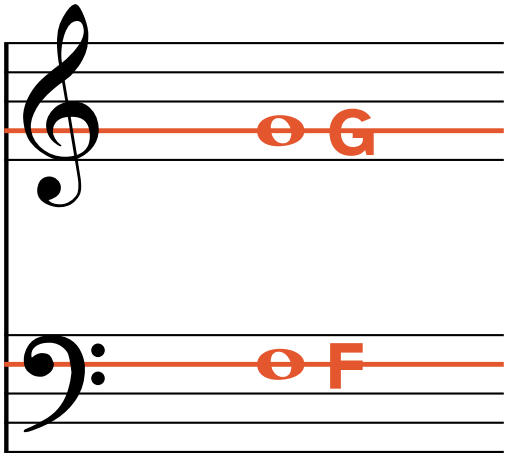
PART I · PITCH · 2

The staff and the clefs

Music is written on a **staff**: five lines and four spaces, counted from the **bottom**. Higher on the staff means higher on the keyboard. The **clef** at the start is the key to the code: it names one line, and every other note follows from it.

THE SECRET

The treble clef curls round G; the bass clef's two dots hug F.





Treble clef
curls round line 2: G

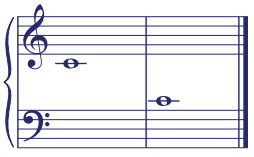


Bass clef
its dots hug line 4: F

Look closely: the treble clef curls round G · the bass clef's dots hug F

The grand staff

Piano music uses two staves joined by a brace: the **treble** staff on top, usually for the right hand, and the **bass** staff below, usually for the left. **Middle C** sits between them on a short line of its own, and can be written in either staff.

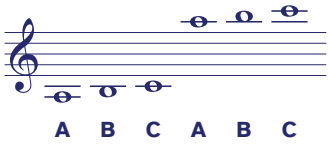


Middle C in the treble staff (left) and in the bass staff (right): the same key.

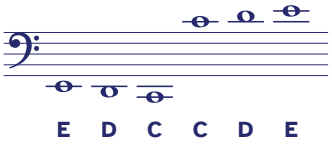
Ledger lines: the staff carried on

Notes too high or too low for the staff sit on short extra lines called **ledger lines**, spaced exactly like the staff lines. Read them by stepping on from the nearest staff line: line, space, line, space.

PART I · PITCH · 2 · CONTINUED



Treble: A, B, C below; A, B, C above.



Bass: E, D, C below; C, D, E above.

8va: play it an octave higher

Lots of ledger lines are hard to read. So a very high passage is written on the staff with **8va** and a dotted line above it: play those notes **one octave higher**, eight notes up, until the dotted line stops. **8vb** under the staff means an octave lower.

THINK OF IT A clef is a pair of glasses: the same five lines read as different notes, depending on which pair you put on.

EXAMPLE 1

One line, two notes

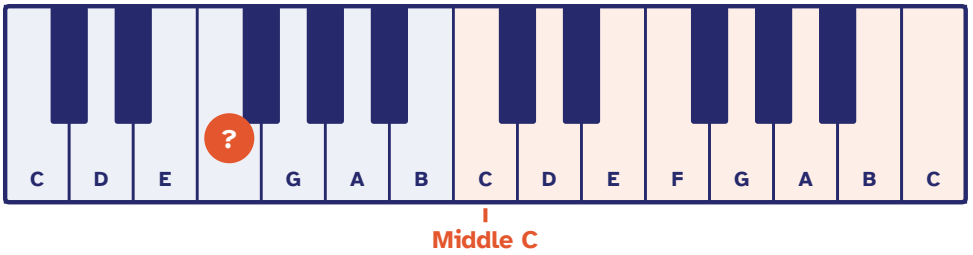


1. **Look at the red notes:** both sit on the bottom line.
2. **Now look at the clefs:** treble on the left, bass on the right.
3. **Read each from its clef note:** treble G is line 2, so step down (F in the space) to line 1; bass F is line 4, so step down line by line.

Treble bottom line: **E**. Bass bottom line: **G**. Same line, different note: always read the **clef** first.

EXAMPLE 2

Which staff is this key written on?



1. **Find Middle C.**
2. **Is the ? to its left or right?** Left is lower: the blue half.
3. **Low keys go on the bass staff**, high keys on the treble staff.

The **?** is below Middle C, so it is written on the **bass staff**, usually for the left hand.

PART I · PITCH · 2 · CONTINUED

EXAMPLE 3

Read a high note on ledger lines



1. **The top line** of the treble staff is F.
2. **Step up** one note at a time: G sits on top of the staff, A on the first ledger line.
3. **Keep stepping** to the red note: B above it, then the second ledger line.

Two ledger lines above the treble staff: **High C**, two octaves above Middle C.

GAME

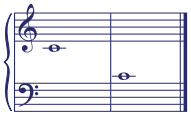
Treble or bass?

1	2	3	4
the E above Middle C	the lowest key on the piano	Middle C	High C

Which staff would each note be written on?

Answer: 1 treble · 2 bass, with ledger lines · 3 either: it sits between the two · 4 treble, two ledger lines up.

TRY IT · 10 SECONDS
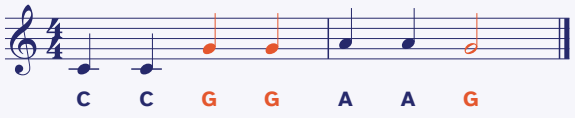


Point, then play.

Point to the line the treble clef curls round and play that G. Point to the line between the bass clef's dots and play that F. Middle C sits between them.

YOU KNOW THIS

Twinkle, Twinkle, Little Star



The treble clef curls round **line 2: G**, the very note Twinkle leaps up to. Middle C hangs on its own ledger line below.

Try it. Point to the G line, then play C C G G A A G.

TRAP

- × Reading bass-clef notes as if they were treble: check the clef on every line.
- × Reading ledger lines from the wrong end: start at the staff and step outwards.

SEE ALSO
Reading the notes p. 29 · The staff and the keyboard p. 32

PRACTISE IT
Workbook Pitch 2 · Online: Library › Staff and clefs

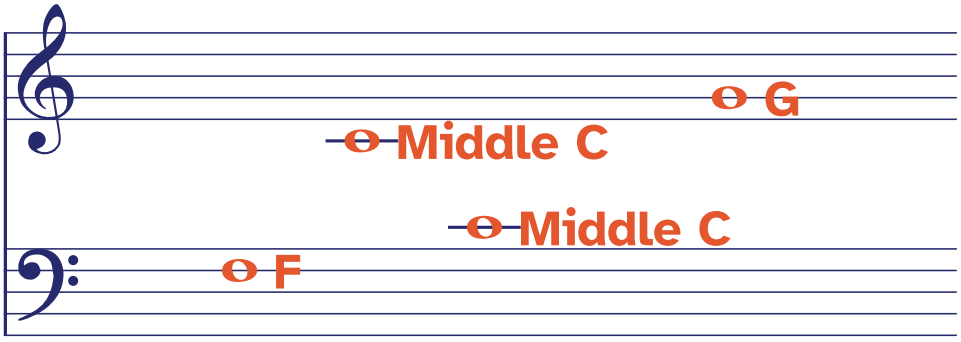
PART I · PITCH · 3

Reading the notes

Every line and space has one letter. You do not have to learn them all at once: start from a note you always know, then **step** or **skip** to the one you want.

THE SECRET

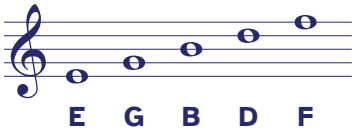
Start from a note you know (bass F, Middle C, treble G), then step or skip.



Notes you always know: treble G, bass F, and Middle C in either clef

THE RULE

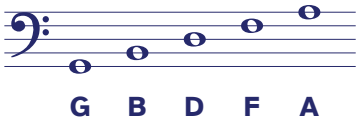
Step: to the very next line or space (one letter on). Skip: line to the next line, or space to the next space (two letters on).



Treble lines, E G B D F: *Every Good Boy Deserves Fun*



Treble spaces, F A C E: they spell *FACE*



Bass lines, G B D F A: *Great Big Dragons Fly Away*

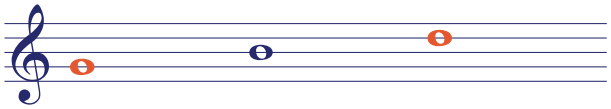


Bass spaces, A C E G: *All Cows Eat Grass*

PART I · PITCH · 3 · CONTINUED

EXAMPLE 1

Read a note you are not sure of

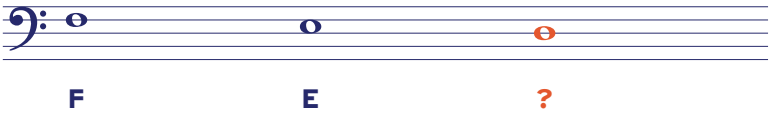


1. **The clef:** treble, so the clef note is G, on line 2.
2. **From G, skip up** line to line: one letter jumped each time.
3. **Count the skips** up to the other red note, on line 4: G, then B, then …

The red note is **D**: G, B, D, two skips up from the clef's G.

EXAMPLE 2

Read a bass note from the clef

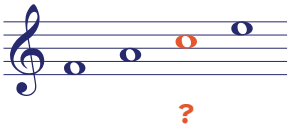


1. **The bass clef's dots** sit either side of line 4: that line is F.
2. **Step down** to the space below.
3. **One more step down** to the red note on line 3.

F, E, **D**: bass line 3 is D, two steps down from the clef's F.

EXAMPLE 3

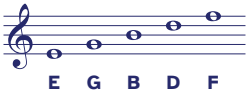
Find Treble C on the staff



1. **These notes sit in the spaces.** The bottom space is F.
2. **Step up through the spaces:** F, A … they spell a word.
3. **The red note is space 3.** Which letter is it?

F A **C** E: space 3 is **C**, Treble C, the same C you found on the keyboard, one octave above Middle C.

TRY IT · 10 SECONDS



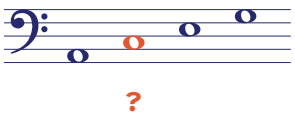
Say the treble lines up and down.

"E G B D F", then "F D B G E", out loud, as fast as you can without a slip. Then the spaces: "F A C E".

PART I · PITCH · 3 · CONTINUED

EXAMPLE 4

Find Bass C on the staff



1. **These notes sit in the bass spaces.** The bottom space is A: *All.*
2. **Say the next word:** *All Cows …*
3. **The red note is space 2.** Which letter?

A **C** E G: space 2 of the bass staff is **C**, Bass C, one octave below Middle C.

GAME

Name it from the clef note
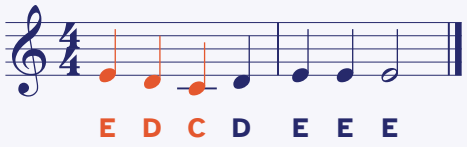
1	2	3	4
treble line 3	treble space 1	bass line 3	bass space 3

Start from the clef note (treble G on line 2, bass F on line 4), then step or skip.

Answer: 1 B (skip up from G) · 2 F (step down from G) · 3 D (two steps down from F) · 4 E (step down from F).

YOU KNOW THIS

Mary Had a Little Lamb



E D C: **line, space, ledger line**, one step down each time. Read the shape and the tune walks down with it.

Try it. Read the notes aloud before you play them.

THINK OF IT The clef notes are lamp posts on a dark road: walk from the nearest one and you are never lost.

TRAP

- × Naming bass notes with treble names: in the bass, the bottom line is G, not E.
- × Mixing up step and skip: a step changes line to space; a skip stays on lines (or on spaces).

SEE ALSO

The staff and the clefs p. 26 · The staff and the keyboard p. 32

PRACTISE IT

Workbook Pitch 2 · Online: Library › Staff and clefs

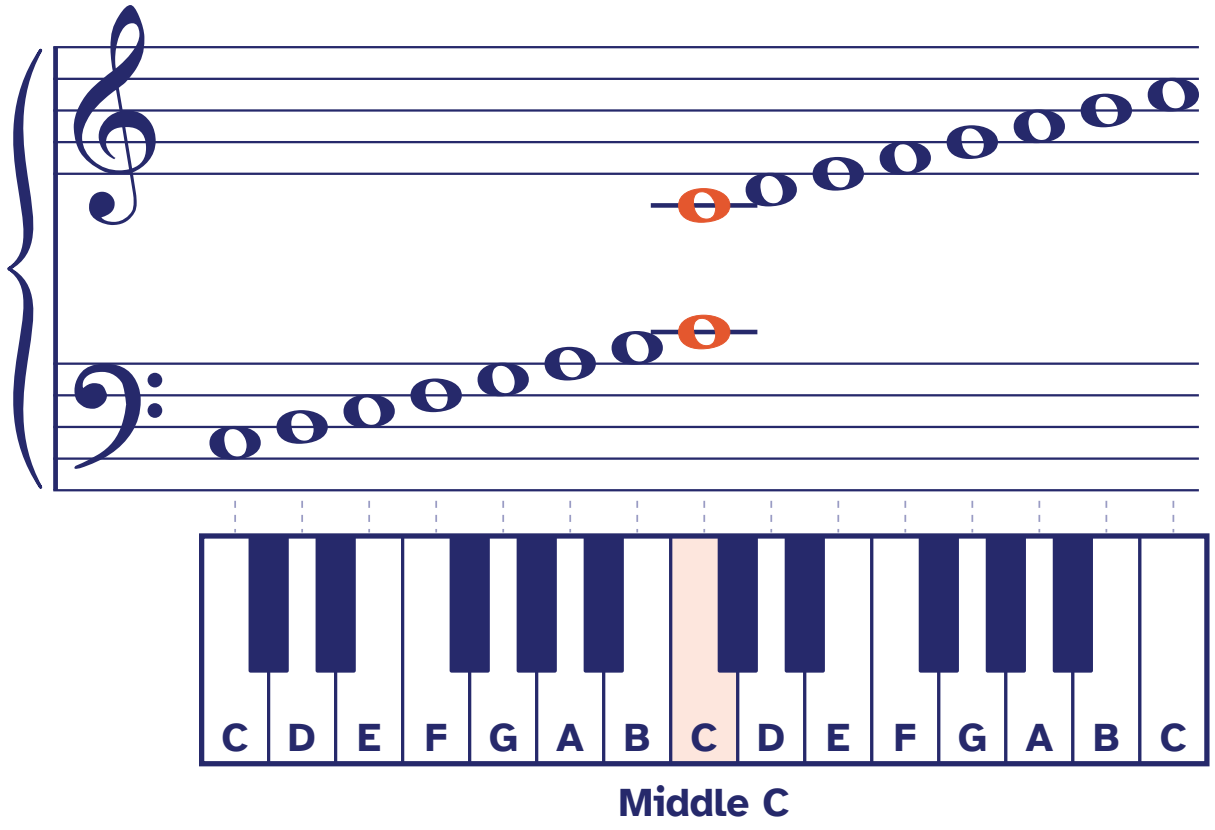
PART I · PITCH · 4

The staff and the keyboard

Each line and each space of the grand staff is one white key. Climb the staff and you climb the keyboard: here are two octaves, from Bass C to Treble C, with Middle C in red where the two staves meet.

THE SECRET

Every line and every space is one white key, in order.



Middle C can be written in either staff: on a ledger line below the treble staff, or above the bass staff. Same key; the hand that plays it chooses.

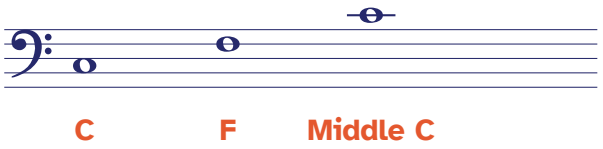
THE RULE

One step on the staff (line to space, or space to line) is the very next white key. The black keys sit in between; how they are written is the next chapter (p. 35).

PART I · PITCH · 4 · CONTINUED

EXAMPLE 1

From the page to the key

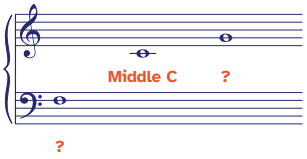


1. **Name the note:** line 4 of the bass staff, between the clef's dots, is F.
2. **On the piano,** find the three black keys just below Middle C.
3. **F is the first key** of that family: play it with the left hand.

The bass-clef F is the **F just below Middle C**: left of the three black keys under it.

EXAMPLE 2

Mirror notes either side of Middle C

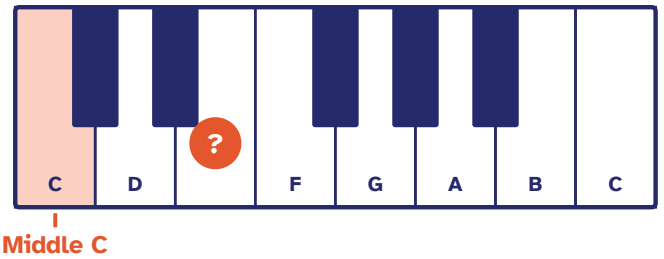


1. **Treble G:** the three black keys just above Middle C; G is the second key, after F.
2. **Bass F:** the three black keys just below Middle C; F is the first key.
3. **Now count each from Middle C:** C D E F G up, C B A G F down. Which number?

Both are **note 5**: treble G and bass F are mirror images, the same distance either side of Middle C.

EXAMPLE 3

From the key to the page



1. **Find the ? key:** count up from Middle C, C 1, D 2 ...
2. **Do the same on the staff:** Middle C on its ledger line, D in the space under the staff ...
3. **Where do you land?**

The **?** is **E**, note 3, and it sits on the **bottom line** of the treble staff.

PART I · PITCH · 4 · CONTINUED

GAME

Count the notes

1 **Middle C up to A**

2 **Middle C down to E**

3 **G up to the next G**

Count the notes, both ends included. The staff climbs the same way: one line or space per note.

Answer: 1 six (C D E F G A) · 2 six (C B A G F E) · 3 eight: an octave.

TRY IT · 10 SECONDS

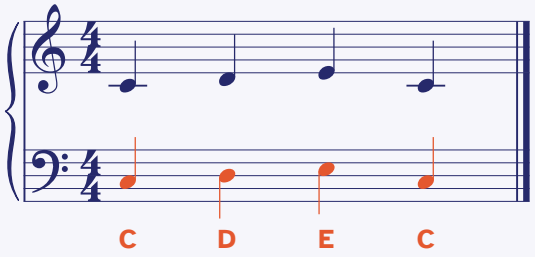


Play Middle C, then walk up.

Say each white key as you play it, C D E F G A B C, and picture each one climbing the staff: line, space, line, space.

YOU KNOW THIS

Frère Jacques, in both hands



The left hand plays the **same letters**, C D E C, one octave lower, written in the bass clef.

Try it. Play it right hand, then left hand, then both together.

THINK OF IT The staff is a ladder laid beside the keyboard: each rung, a line or a space, is the next white key.

TRAP

- × Skipping a key: every line and every space is used, one white key each.
- × Counting from 0: the note you start on is note 1.

SEE ALSO

The keyboard p. 22 · Reading the notes p. 29

PRACTISE IT

Workbook Pitch 1–2 · Online: Library › Staff and clefs

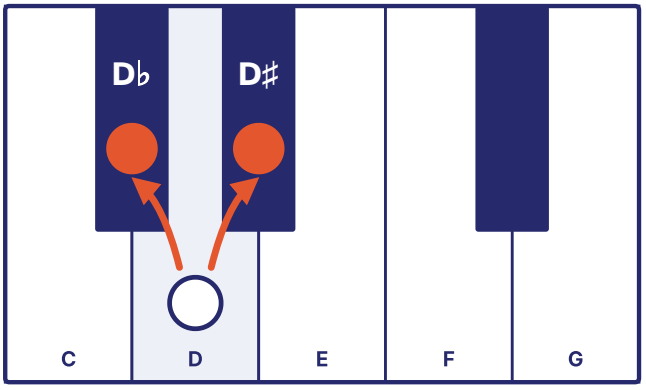
PART I · PITCH · 5

Sharps, flats and naturals

A sign in front of a note moves it to the very next key. **Sharp**: one key to the **right**. **Flat**: one key to the **left**. Usually that next key is black, which is how the black keys get their names.

THE SECRET

Sharp: the very next key to the right. Flat: the very next key to the left.

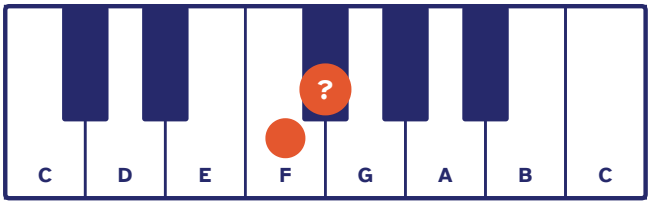


♯ one key right · ♭ one key left

SIGN	NAME	WHAT IT DOES	FIRST MET
♯	sharp	one key to the right (a semitone higher)	Level 0
♭	flat	one key to the left (a semitone lower)	Level 0
♮	natural	cancels a sharp or flat: back to the white key	Level ½

EXAMPLE 1

Find F♯



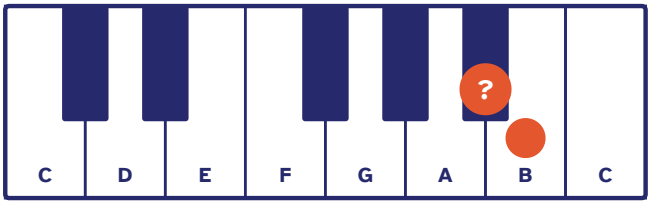
1. **Find F** (the red dot): just left of the three black keys.
2. **Sharp means right:** move to the very next key.
3. **Is it black or white?** Put your finger on it.

The **?** is **F♯**: the first black key of the group of three.

PART I · PITCH · 5 · CONTINUED

EXAMPLE 2

Find B♭
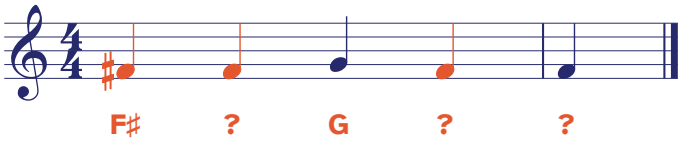


1. **Find B** (the red dot): just right of the three black keys.
2. **Flat means left:** move to the very next key.
3. **Which black key is it?**

The **?** is **B♭**: the top black key of the group of three.

EXAMPLE 3

How long does the sharp last?

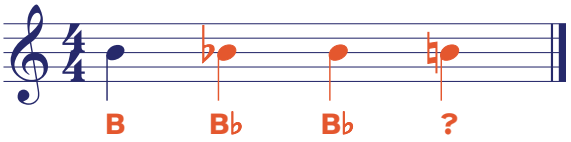


1. **The first note** has a sharp: F♯.
2. **Later in the same bar** there is no sign. Is it still sharp?
3. **After the bar line**: is it still sharp?

Still **F♯** until the bar line; after it, plain **F** again. A sign lasts **to the end of its bar**.

EXAMPLE 4

A natural cancels the flat
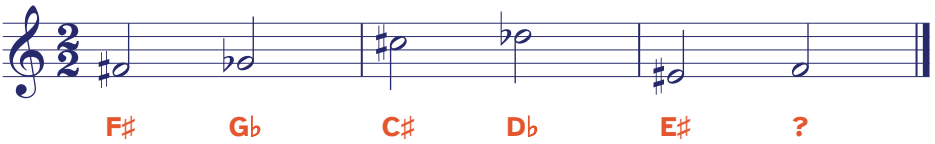


1. **B**: the white key.
2. **B♭**: one key left, and it stays flat for the rest of the bar.
3. **The last note** has a natural sign: what does it do?

The natural **cancels** the flat: back to the white key B, until the bar line.

PART I · PITCH · 5 · CONTINUED

EXAMPLE 5

One key, two names



1 **The first bar:** F♯, then G♭. Find both on the keyboard.

2 **Are they different keys?**

3 **The last bar:** E♯. Which key is one right of E?

F♯ and G♭ are the **same black key**; E♯ is the white key **F**. Same sound, different names: **enharmonic**.

GAME

Which key is it?
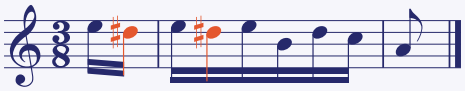
1	2	3	4
C♯	E♭	B♯	F♭

Sharp: one key to the right. Flat: one key to the left. Which key do you press?

Answer: 1 the black key right of C · 2 the black key left of E · 3 the white key C · 4 the white key E.

YOU KNOW THIS

Für Elise (Beethoven)



E, D♯, E, D♯: the very next key down and back. The sharp is written again in the new bar, because the bar line wiped it out.

Try it. Play E and D♯ back and forth: the black key just left of E.

THINK OF IT A sharp or flat is a sticky note on one line of the staff: it stays stuck until the bar line, then it falls off.

TRAP

× Carrying a sharp or flat past the bar line: after it, the note is plain again.

SEE ALSO
Keys and key signatures p. 76

PRACTISE IT
Workbook Pitch 3 · Online: Library › Signs

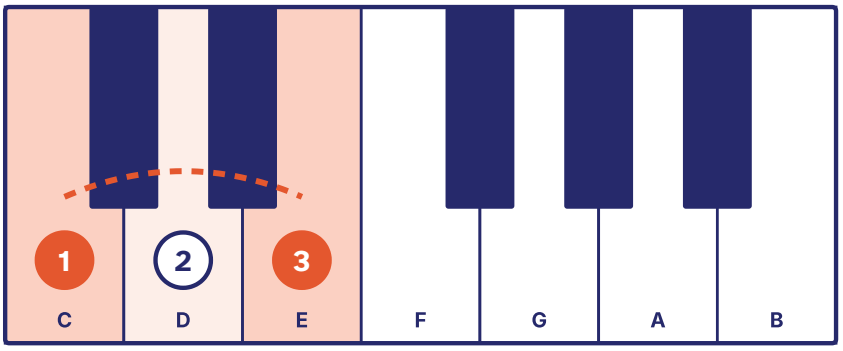
PART I · PITCH · 6

Counting intervals

How far apart are two notes? That distance is an **interval**. To give it a number, **count the notes**: the first note is 1, and you count every letter up to the second, both ends included.

THE SECRET

Count the notes, both ends included: C D E is 3 notes, a 3rd.

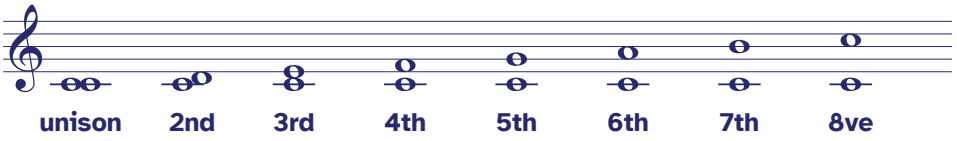


C to E: count the notes, 1 2 3: a 3rd

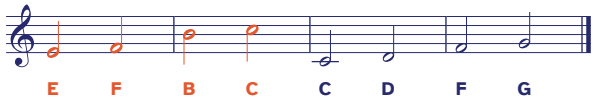
THE RULE

Count the notes for the number: start at 1 on the first note. C to E is C D E: 3 notes, a 3rd. Sharps and flats never change the number.

Interval numbers from C



The smallest steps: semitones and tones



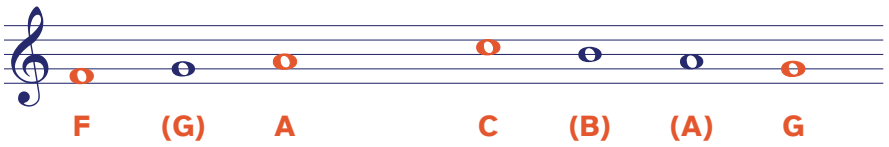
E–F and B–C: no black key between, a semitone. C–D and F–G: a black key between, a tone.

From any key to the **very next key**, black or white, is a **semitone**: the smallest step on the piano. Two semitones make a **tone**. On the white keys most steps are tones, but **E–F** and **B–C** have no black key between them: they are semitones.

PART I · PITCH · 6 · CONTINUED

EXAMPLE 1

How far apart are these notes?

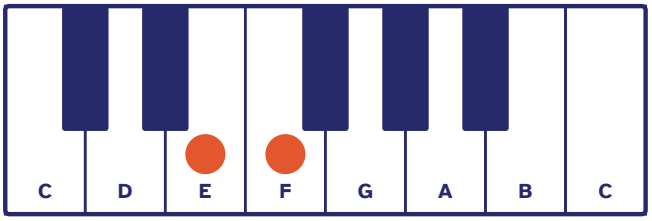


1. **Call the first note 1:** here F is 1.
2. **Count every letter** up to the second note.
3. **Going down works the same:** start at 1 on the top note and count down.

F G A is 3 notes: a **3rd**. C B A G is 4 notes: a **4th**.

EXAMPLE 2

Tone or semitone?

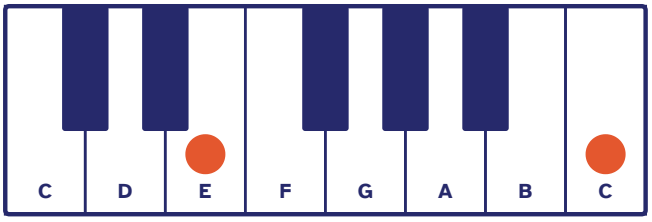


1. **Look between the red keys:** is there a black key?
2. **No black key** means they are next-door keys.
3. **Now try C and D**: is there a black key between?

E–F is a **semitone** (next door). C–D has C♯ between: a **tone**.

EXAMPLE 3

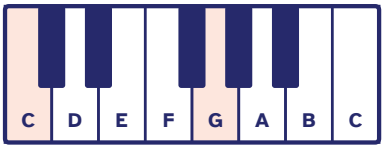
Name an interval on the keyboard



1. **Find both red keys:** E, and the C above it.
2. **Count the notes** from E: E is 1 …
3. **Stop on the C.** What number is it?

E F G A B C is 6 notes: a **6th**.

TRY IT · 10 SECONDS



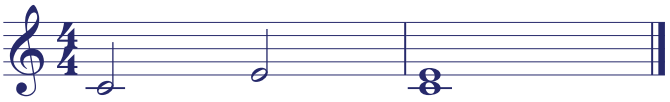
Thumb on C, finger 5 on G.

Your hand is a ruler: thumb to little finger on white keys is a 5th. Play it, then thumb to finger 3: a 3rd.

PART I · PITCH · 6 · CONTINUED

EXAMPLE 4

Melodic or harmonic?



1 **The first bar:** C, then E, one after the other.

2 **The second bar:** C and E at the same time.

3 **Count the notes** in each.

Both are a **3rd**: **melodic** when played one after the other, **harmonic** when played together.

GAME

Count the notes
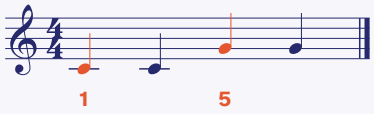
1	2	3	4
C up to G	D up to F	E up to the next E	G up to A

Count every note, both ends included. What is each interval called?

Answer: 1 a 5th (C D E F G) · 2 a 3rd · 3 an octave (8) · 4 a 2nd.

YOU KNOW THIS

Twinkle, Twinkle, Little Star



The leap from the first “twinkle” to the second is C up to G: C D E F G, **five notes, a 5th**.

Try it. Sing the first two notes of any song, then count the notes between them.

THINK OF IT Monday to Wednesday is three days, not two: you count the day you start on. Intervals count the same way.

TRAP

- × Counting only the gaps, not the notes: C to E is 3 notes, not 2 gaps.
- × Letting a sharp or flat change the number: C to G♯ is still a 5th.

SEE ALSO

Major, minor and perfect p. 41 · Major scales p. 70

PRACTISE IT

Workbook Pitch 4 · Online: Library › Intervals

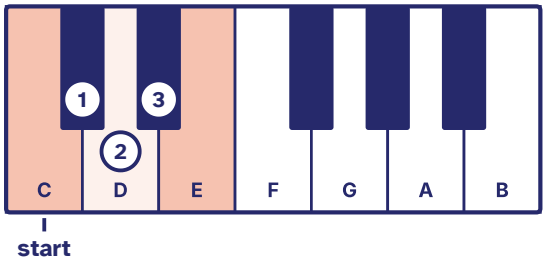
PART I · PITCH · 7

Major, minor and perfect

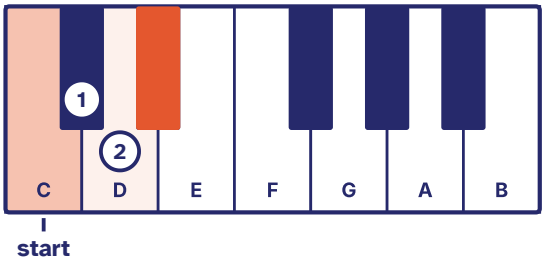
C to E and C to E♭ are both 3rds: three notes each. But they sound different, bright and dark. To tell them apart, **count the keys in between**, black and white: the keys you do not play.

THE SECRET

Count the keys between: 3 make a major 3rd, 2 a minor 3rd.



C to **E**: 3 keys between. A **major** 3rd: bright.



C to **E♭**: 2 keys between, one fewer. A **minor** 3rd: dark.

One key fewer between: a major 3rd becomes minor

THE RULE

Count the notes for the number (a 3rd). Count the keys in between for the size (major or minor): black and white, not the two you play.

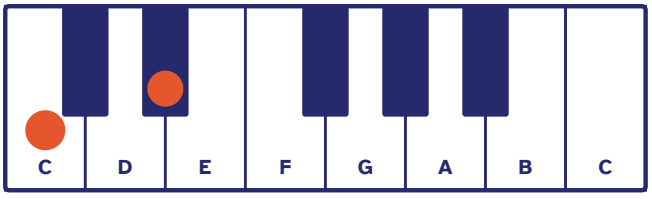
NUMBER	KEYS BETWEEN: SMALLER	MAIN SIZE
3rd	2 (minor)	3 (major)
4th		4 (perfect)
5th		6 (perfect)
6th	7 (minor)	8 (major)
octave		11 (perfect)

3rds and 6ths come in two sizes: **major**, and **minor**, one key fewer between.

4ths, 5ths and octaves have one main size, called **perfect**. They are never major or minor.

EXAMPLE 1

Major or minor 3rd?



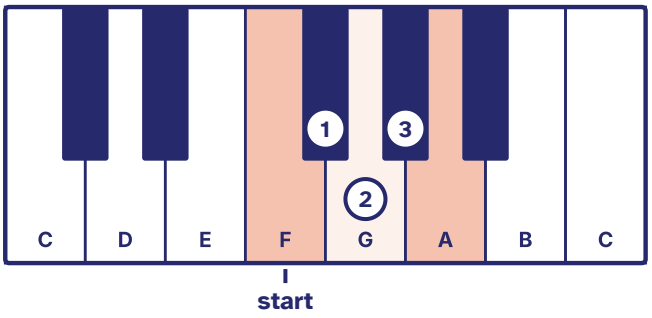
1. **Count the notes:** C D E, three letters, so it is a 3rd.
2. **Count the keys in between** C and the red black key, E♭: C♯, D …
3. **A major 3rd has 3 between.** How many here?

C to E♭ has **2 keys between**: a **minor 3rd**.

PART I · PITCH · 7 · CONTINUED

EXAMPLE 2

Name F up to A

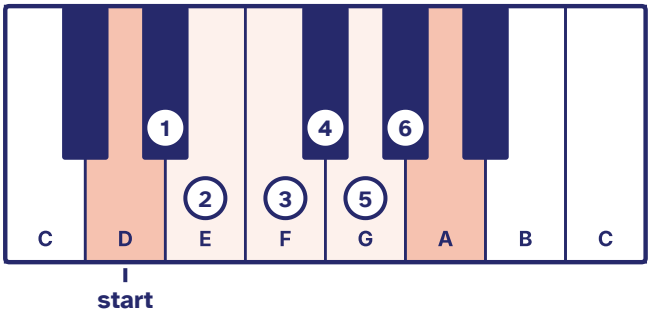


1. **Count the notes:** F G A is 3, so it is a 3rd.
2. **Count the keys in between:** they are numbered.
3. **3 between, or 2?**

F to A has 3 keys between: a **major 3rd**, the same size as C to E.

EXAMPLE 3

Name D up to A



1. **The number:** D E F G A, five notes, a 5th.
2. **The size:** count the keys in between: the numbered keys.
3. **A 5th with 6 between** is its main size. What is that called?

D up to A is a **perfect 5th**: 6 keys between.

GAME

Perfect, major or minor?

1	2	3	4
C–G	D–F	C–E	E–A

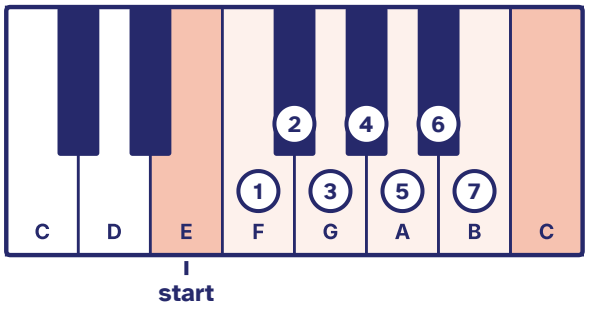
Count the notes for the number, then the keys in between for the size.

Answer: 1 perfect 5th (6 between) · 2 minor 3rd (2) · 3 major 3rd (3) · 4 perfect 4th (4).

PART I · PITCH · 7 · CONTINUED

EXAMPLE 4

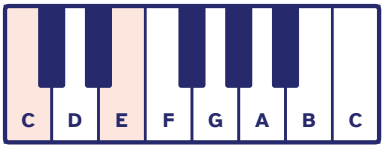
Name E up to C



1. **The number:** E F G A B C, six notes, a 6th.
2. **The size:** count the keys in between.
3. **A major 6th** has 8 between. Is this one more or one fewer?

E up to C has 7 keys between, one fewer than major: a **minor 6th**.

TRY IT · 10 SECONDS
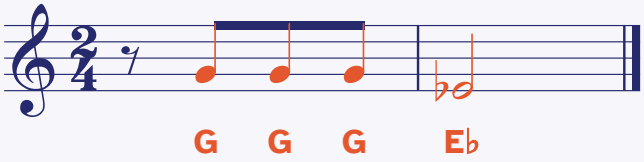


Play C–E, then C–E♭.

Bright, then dark. One key makes the difference: the major 3rd has three keys between, the minor 3rd two.

YOU KNOW THIS

Beethoven, Symphony No. 5



Da-da-da-DUM: G down to E♭ is three notes, a 3rd, with 3 keys between (F♯, F, E): a **major 3rd**.

Try it. Count the keys between G and E♭ on the keyboard.

THINK OF IT A major and a minor 3rd are a smile and a frown: the same face, one small change at the corners.

TRAP

- × Counting white keys only: count every key between, black and white.
- × Counting the two notes you play: count only the keys in between.
- × Calling a 4th or 5th “major”: they are perfect.

SEE ALSO
Counting intervals p. 38 · Chords and triads p. 87

PRACTISE IT
Workbook Pitch 4 · Online: Library › Intervals

PART I · PITCH · TIPS

When a note reads wrong

Most wrong notes come from a handful of slips. Name the slip first, then fix only that note: playing the whole line again rarely finds it.

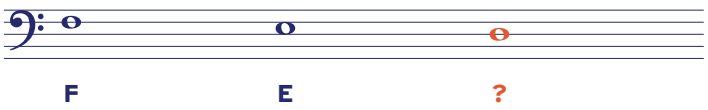
TIP

What you notice, what to check

WHAT YOU NOTICE	CHECK	TURN TO
The left hand reads like the right	**the clef**: bass lines are G B D F A, not E G B D F	p. 26
You land an octave too high or too low	**the Cs**: find Middle C first, then walk out to Bass C or Treble C	p. 22
Notes above or below the staff slow you down	**the ledger lines**: step on from the nearest staff line	p. 26
A sharp or flat vanishes, or stays too long	**the bar line**: a sign lasts only to the end of its bar	p. 36
You play the black key on the wrong side	**the direction**: sharp goes right, flat goes left	p. 35
Two notes sound too close or too far apart	**the interval**: count the notes, then the keys between	p. 41

EXAMPLE

Spot the slip



Sam played B. Wrong!

1. **Check the clef first:** this is the bass clef, not the treble.
2. **Start from the clef note:** the dots hug line 4, F.
3. **Step down** to the red note on line 3: F, E …

The note is **D**. Sam read the bass staff with treble names (treble line 3 is B): the most common slip of all.

HOW TO

Three checks before you read a new line

1. **The clefs:** which staff is treble, which is bass?
2. **The clef notes:** find treble G, bass F and Middle C, and read the other notes from them.
3. **The signs:** mark each sharp, flat or natural, and see where its bar ends.

THINK OF IT Like reading a map: find the big landmarks first, then the streets around them. A note far from every clef note is the one most often misread.